Delivery - On The Go: Data Analyst Technical Assessment

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Executive Summary

Following a +21.9% citywide delivery attempt surge in Week 2, hub managers submitted a blanket request to hire 8 new delivery agents. Management must REJECT this blanket request. Approving 8 new hires incurs BDT 175,200 in Month 1 (BDT 40,000 in recruitment fees + BDT 135,200 in fixed salaries), exceeding the BDT 70,000 Finance budget by 2.5x. Furthermore, new hires operate at only 60% speed during their first 7 days, providing zero immediate relief to ongoing bottlenecks.

Empirical audit of 51,190 delivery logs reveals that capacity strain is highly asymmetric. Only 3 of 6 hubs exceed the 85% planning threshold in Week 2: Greenfield H01 (88.64%), Riverside H03 (90.80%), and Hillcrest H04 (95.87%). Overload at Riverside (H03) is largely artificial: 64.42% of its holds (563 holds) are caused by internal agent-side failures (Late Dispatch, Not Attempted, Time Shortage) that recycle into repeat next-day attempts, draining BDT 37,000 in penalties in Week 2 alone. Meanwhile, Lakeview (H02) in Cluster A operates at just 56.17% utilisation (<75% transfer threshold), holding 4 surplus agents directly adjacent to Greenfield.

We recommend an optimized 4-part plan: (1) Transfer 2 agents from Lakeview (H02) to Greenfield (H01) within Cluster A at BDT 2,000 each (BDT 4,000 total); (2) Deploy a BDT 15,000 Hold Reduction Programme at Riverside (H03) to eliminate 50% of agent-side holds (~282 holds/wk), reducing required staff to 11.6 agents (already has 12 active); (3) Approve targeted hiring of 2 agents at Hillcrest (H04) where genuine demand growth exists and no intra-cluster donor is available (BDT 46,698 total); and (4) Coach 3 senior return outliers at Brookside (H05) to eliminate BDT 14,135 in return penalties. Total expenditure is strictly BDT 65,698 (leaving BDT 4,302 contingency). This unlocks BDT 35,695 in direct cash penalty savings and protects BDT 111,461 in revenue, delivering a +110.2% Net ROI.

Task 1: SQL Assessment Queries & Result Tables

Q1. Daily Agent Productivity

Every delivery attempt is initiated by an Assigned status event in `order_logs`, followed by exactly one outcome (Delivered, Hold, or Returned) on the same date. Inactive agents departed at the end of Week 1. We calculate Deliveries per Working Agent-Day (days with activity) and Deliveries per Scheduled Day (882 available days).

```sql
-- ============================================================================
-- DELIVERY - ON THE GO: DATA ANALYST TECHNICAL ASSESSMENT
-- TASK 1: SQL ASSESSMENT QUERIES (Q1 - Q7)
-- Database: delivery_assessment.db (SQLite)
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-- ============================================================================

-- ----------------------------------------------------------------------------
-- Q1. Daily Agent Productivity
-- ----------------------------------------------------------------------------
-- Business Context & Logic:
-- Every delivery attempt initiates with an 'Assigned' status event in `order_logs`.
-- On the same day, exactly one terminal outcome occurs: 'Delivered', 'Hold', or 'Returned'.
-- Inactive agents (A1011 in H01, A1064 in H06) left the company at the end of Week 1
-- and therefore contributed 6 working days in Week 1 and 0 in Week 2.
-- We compute two standard industry benchmarks for hub average deliveries per agent-day:
-- 1) Average Deliveries per Working Agent-Day: Deliveries / Actual Agent-Days with logged activity.
-- 2) Average Deliveries per Available Agent-Day: Deliveries / Total Scheduled Agent-Days (64*7 in W1, 62*7 in W2).

-- Part A: Daily Agent Productivity Table (Top 20 rows presented for submission)
WITH daily_agent_activity AS (
    SELECT
        DATE(ol.event_time) AS work_date,
        ol.agent_id,
        a.agent_name,
        a.hub_id,
        h.hub_name,
        h.cluster,
        a.salary_type,
        a.active_status,
        SUM(CASE WHEN ol.status = 'Assigned' THEN 1 ELSE 0 END) AS assigned_attempts,
        SUM(CASE WHEN ol.status = 'Delivered' THEN 1 ELSE 0 END) AS delivered_orders,
        SUM(CASE WHEN ol.status = 'Hold' THEN 1 ELSE 0 END) AS hold_orders,
        SUM(CASE WHEN ol.status = 'Returned' THEN 1 ELSE 0 END) AS returned_orders
    FROM order_logs ol
    JOIN agents a ON ol.agent_id = a.agent_id
    JOIN hubs h ON a.hub_id = h.hub_id
    GROUP BY
        DATE(ol.event_time),
        ol.agent_id,
        a.agent_name,
        a.hub_id,
        h.hub_name,
        h.cluster,
        a.salary_type,
        a.active_status
)
SELECT
    work_date,
    agent_id,
    agent_name,
    hub_id,
    hub_name,
    cluster,
    salary_type,
    active_status,
    assigned_attempts,
    delivered_orders,
    hold_orders,
    returned_orders
FROM daily_agent_activity
ORDER BY work_date ASC, hub_id ASC, agent_id ASC;


-- Part B: Full Hub Summary Table (6 Hubs)
WITH daily_agent_activity AS (
    SELECT
        DATE(ol.event_time) AS work_date,
        ol.agent_id,
        a.hub_id,
        SUM(CASE WHEN ol.status = 'Assigned' THEN 1 ELSE 0 END) AS assigned_attempts,
        SUM(CASE WHEN ol.status = 'Delivered' THEN 1 ELSE 0 END) AS delivered_orders,
        SUM(CASE WHEN ol.status = 'Hold' THEN 1 ELSE 0 END) AS hold_orders,
        SUM(CASE WHEN ol.status = 'Returned' THEN 1 ELSE 0 END) AS returned_orders
    FROM order_logs ol
    JOIN agents a ON ol.agent_id = a.agent_id
    GROUP BY DATE(ol.event_time), ol.agent_id, a.hub_id
),
hub_working_metrics AS (
    SELECT
        hub_id,
        COUNT(DISTINCT agent_id) AS total_agents_rostered,
        COUNT(*) AS total_working_agent_days,
        SUM(assigned_attempts) AS total_assigned_attempts,
        SUM(delivered_orders) AS total_delivered_orders,
        SUM(hold_orders) AS total_hold_orders,
        SUM(returned_orders) AS total_returned_orders,
        ROUND(1.0 * SUM(delivered_orders) / COUNT(*), 2) AS avg_deliveries_per_working_day
    FROM daily_agent_activity
    GROUP BY hub_id
),
hub_available_metrics AS (
    SELECT
        h.hub_id,
        -- Week 1: All agents * 7 days; Week 2: Active agents only * 7 days
        (SELECT COUNT(*) * 7 FROM agents a1 WHERE a1.hub_id = h.hub_id) +
        (SELECT COUNT(*) * 7 FROM agents a2 WHERE a2.hub_id = h.hub_id AND a2.active_status = 'Active') AS total_available_agent_days
    FROM hubs h
)
SELECT
    h.hub_id,
    h.hub_name,
    h.cluster,
    ham.total_available_agent_days,
    hwm.total_working_agent_days,
    hwm.total_assigned_attempts,
    hwm.total_delivered_orders,
    hwm.total_hold_orders,
    hwm.total_returned_orders,
    ROUND(1.0 * hwm.total_delivered_orders / ham.total_available_agent_days, 2) AS avg_deliveries_per_available_day,
```

```
    hwm.avg_deliveries_per_working_day
FROM hubs h
JOIN hub_working_metrics hwm ON h.hub_id = hwm.hub_id
JOIN hub_available_metrics ham ON h.hub_id = ham.hub_id
ORDER BY h.hub_id ASC;


-- ------------------------------------------------------------------------
```

Table 1.1: 6-Hub Productivity Summary Table

HUB_ID	HUB_NAME	CLUSTER	TOTAL_AVAILABLE_AGENT_DAYS	TOTAL_WORKING_AGENT_DAYS	TOTAL_ASSIGNED_ATTEMPTS	TOTAL_DELIVERED_ORDERS	TOTAL_HOLD_ORDERS	TOTAL_RETURNED_ORDERS	AVG
H01	Greenfield	Cluster A	147	144	4407	3908	327	172	26.5
H02	Lakeview	Cluster A	154	149	3377	3001	244	132	19.4
H03	Riverside	Cluster B	168	165	5332	3479	1501	352	20.7
H04	Hillcrest	Cluster B	126	123	4359	3798	343	218	30.1
H05	Brookside	Cluster C	154	147	4225	3411	357	457	22.1
H06	Westwood	Cluster C	133	126	3895	3481	270	144	26.1

Table 1.2: First 20 Rows of Daily Agent Productivity Table

WORK_DATE	AGENT_ID	AGENT_NAME	HUB_ID	HUB_NAME	CLUSTER	SALARY_TYPE	ACTIVE_STATUS	ASSIGNED_ATTEMPTS	DELIVERED_ORDERS	HOLD_ORDERS	RETURNED_ORDERS
2026-10-13	A1001	Weno Tremaine	H01	Greenfield	Cluster A	Variable	Active	20	18	2	0
2026-10-13	A1002	Valel Wexley	H01	Greenfield	Cluster A	Fixed	Active	21	17	4	0
2026-10-13	A1003	Torian Vantell	H01	Greenfield	Cluster A	Fixed	Active	27	22	3	2
2026-10-13	A1004	Garian Larkin	H01	Greenfield	Cluster A	Variable	Active	20	19	1	0
2026-10-13	A1005	Wenan Brannick	H01	Greenfield	Cluster A	Fixed	Active	23	20	2	1
2026-10-13	A1006	Toren Rowntree	H01	Greenfield	Cluster A	Fixed	Active	26	23	2	1
2026-10-13	A1007	Coris Rowntree	H01	Greenfield	Cluster A	Fixed	Active	27	25	0	2
2026-10-13	A1008	Garon Holloway	H01	Greenfield	Cluster A	Variable	Active	26	22	3	1
2026-10-13	A1009	Perian Halvorsen	H01	Greenfield	Cluster A	Variable	Active	34	27	6	1
2026-10-13	A1010	Joris Fenwick	H01	Greenfield	Cluster A	Fixed	Active	28	23	2	3
2026-10-13	A1011	Marel Fenwick	H01	Greenfield	Cluster A	Fixed	Inactive	14	12	1	1
2026-10-13	A1012	Zarik Rowntree	H02	Lakeview	Cluster A	Fixed	Active	19	18	1	0
2026-10-13	A1013	Pero Ketteridge	H02	Lakeview	Cluster A	Fixed	Active	25	22	2	1
2026-10-13	A1014	Zaris Quorne	H02	Lakeview	Cluster A	Fixed	Active	26	23	2	1
2026-10-13	A1015	Toren Larkin	H02	Lakeview	Cluster A	Fixed	Active	24	21	3	0
2026-10-13	A1016	Noren Vantell	H02	Lakeview	Cluster A	Fixed	Active	17	15	0	2
2026-10-13	A1017	Joran Pelham	H02	Lakeview	Cluster A	Variable	Active	23	23	0	0
2026-10-13	A1018	Lorus Brannick	H02	Lakeview	Cluster A	Fixed	Active	14	13	1	0
2026-10-13	A1019	Selus Rowntree	H02	Lakeview	Cluster A	Fixed	Active	19	17	2	0
2026-10-13	A1020	Noris Fenwick	H02	Lakeview	Cluster A	Variable	Active	15	14	1	0

Q2. Hub Utilisation in Week 1 and Week 2

Planned Utilisation = Total Attempts / (Available Active Agents * 7 days * 40 standard attempts/day). The planning target is 85.0% (34 attempts/day). Flags identify Overloaded (>85%), Eligible Donor (<75%), and Balanced hubs.

```
Hub Utilisation in Week 1 and Week 2
-- ------------------------------------------------------------------------
-- Definition of Utilisation:
-- As prescribed in `rules` tab of `cost_sheet.xlsx`:
-- Standard workload = 40 attempts per agent per day.
-- Hubs operate every calendar day (7 operating days per week).
-- Inactive agents left at the end of Week 1 (available in W1, unavailable in W2).
-- Standard Planned Capacity = (Available Active Agents in Week) * 7 days * 40 attempts/day.
-- Planned Utilisation = Total Attempts / Standard Planned Capacity.
-- We also report Actual Worked Utilisation = Total Attempts / (Working Agent-Days * 40).
-- Overload Flag: Triggered when Planned Utilisation exceeds the 85% planning target (0.85).

WITH w1_attempts AS (
    SELECT
        a.hub_id,
        COUNT(DISTINCT DATE(ol.event_time) || '_' || ol.agent_id) AS w1_working_days,
        SUM(CASE WHEN ol.status = 'Assigned' THEN 1 ELSE 0 END) AS w1_total_attempts
    FROM order_logs ol
    JOIN agents a ON ol.agent_id = a.agent_id
    WHERE DATE(ol.event_time) BETWEEN '2026-10-13' AND '2026-10-19'
    GROUP BY a.hub_id
),
w2_attempts AS (
    SELECT
        a.hub_id,
        COUNT(DISTINCT DATE(ol.event_time) || '_' || ol.agent_id) AS w2_working_days,
        SUM(CASE WHEN ol.status = 'Assigned' THEN 1 ELSE 0 END) AS w2_total_attempts
    FROM order_logs ol
    JOIN agents a ON ol.agent_id = a.agent_id
    WHERE DATE(ol.event_time) BETWEEN '2026-10-20' AND '2026-10-26'
    GROUP BY a.hub_id
),
hub_headcounts AS (
    SELECT
        h.hub_id,
        h.hub_name,
        h.cluster,
        COUNT(a.agent_id) AS w1_available_agents,
        SUM(CASE WHEN a.active_status = 'Active' THEN 1 ELSE 0 END) AS w2_available_agents
    FROM hubs h
    JOIN agents a ON h.hub_id = a.hub_id
    GROUP BY h.hub_id, h.hub_name, h.cluster
)
SELECT
    hh.hub_id,
    hh.hub_name,
    hh.cluster,
    -- Week 1 Metrics
    hh.w1_available_agents,
    w1.w1_working_days,
    w1.w1_total_attempts,
    ROUND(1.0 * w1.w1_total_attempts / (hh.w1_available_agents * 7 * 40), 4) AS w1_planned_utilisation,
    ROUND(1.0 * w1.w1_total_attempts / (w1.w1_working_days * 40), 4) AS w1_worked_utilisation,
    -- Week 2 Metrics
    hh.w2_available_agents,
    w2.w2_working_days,
    w2.w2_total_attempts,
    ROUND(1.0 * w2.w2_total_attempts / (hh.w2_available_agents * 7 * 40), 4) AS w2_planned_utilisation,
    ROUND(1.0 * w2.w2_total_attempts / (w2.w2_working_days * 40), 4) AS w2_worked_utilisation,
    -- Evaluation & Flags
    ROUND(100.0 * (w2.w2_total_attempts - w1.w1_total_attempts) / w1.w1_total_attempts, 2) AS attempt_growth_pct,
    CASE
        WHEN (1.0 * w2.w2_total_attempts / (hh.w2_available_agents * 7 * 40)) > 0.85 THEN 'YES (Overloaded > 85%)'
        ELSE 'NO (Within Target)'
    END AS w2_over_target_flag
FROM hub_headcounts hh
JOIN w1_attempts w1 ON hh.hub_id = w1.hub_id
JOIN w2_attempts w2 ON hh.hub_id = w2.hub_id
ORDER BY hh.hub_id ASC;


-- ------------------------------------------------------------------------
```

Table 1.3: Hub Utilisation Comparison (Week 1 vs Week 2)

HUB_ID	HUB_NAME	CLUSTER	W1_AVAILABLE_AGENTS	W1_WORKING_DAYS	W1_TOTAL_ATTEMPTS	W1_PLANNED_UTILISATION	W1_WORKED_UTILISATION	W2_AVAILABLE_AGENTS	W2_WORKING_DAYS	W2
H01	Greenfield	Cluster A	11	75	1925	0.6250	0.6417	10	69	248
H02	Lakeview	Cluster A	11	74	1647	0.5347	0.5564	11	75	173
H03	Riverside	Cluster B	12	83	2281	0.6789	0.6870	12	82	305
H04	Hillcrest	Cluster B	9	62	1943	0.7710	0.7835	9	61	241
H05	Brookside	Cluster C	11	72	2023	0.6568	0.7024	11	75	220
H06	Westwood	Cluster C	10	67	1870	0.6679	0.6978	9	59	202

Q3. Hold Reasons in Week 2

Agent-side hold reasons: Late Dispatch, Not Attempted, and Agent Time Shortage. All other reasons are customer-side.

```
Hold Reasons in Week 2
-- ------------------------------------------------------------------------------
-- Classification Logic:
-- Per Case Notes: 'Late Dispatch', 'Not Attempted', and 'Agent Time Shortage' are agent-side hold reasons.
-- All other hold reasons ('Customer Unreachable', 'Customer Rescheduled', 'Address Incomplete') are customer-side.
-- Period: Week 2 = 2026-10-20 to 2026-10-26.

WITH w2_holds AS (
    SELECT
        a.hub_id,
        h.hub_name,
        h.cluster,
        ol.hold_reason,
        CASE
            WHEN ol.hold_reason IN ('Late Dispatch', 'Not Attempted', 'Agent Time Shortage') THEN 'Agent-side'
            ELSE 'Customer-side'
        END AS reason_category
    FROM order_logs ol
    JOIN agents a ON ol.agent_id = a.agent_id
    JOIN hubs h ON a.hub_id = h.hub_id
    WHERE ol.status = 'Hold'
      AND DATE(ol.event_time) BETWEEN '2026-10-20' AND '2026-10-26'
),
hub_aggregates AS (
    SELECT
        hub_id,
        hub_name,
        cluster,
        COUNT(*) AS total_w2_holds,
        SUM(CASE WHEN reason_category = 'Agent-side' THEN 1 ELSE 0 END) AS agent_side_holds,
        SUM(CASE WHEN reason_category = 'Customer-side' THEN 1 ELSE 0 END) AS customer_side_holds,
        ROUND(100.0 * SUM(CASE WHEN reason_category = 'Agent-side' THEN 1 ELSE 0 END) / COUNT(*), 2) AS agent_side_share_pct
    FROM w2_holds
    GROUP BY hub_id, hub_name, cluster
),
ranked_reasons AS (
    SELECT
        hub_id,
        hold_reason,
        COUNT(*) AS reason_count,
        ROW_NUMBER() OVER (PARTITION BY hub_id ORDER BY COUNT(*) DESC, hold_reason ASC) AS rnk
    FROM w2_holds
    GROUP BY hub_id, hold_reason
)
SELECT
    ha.hub_id,
    ha.hub_name,
    ha.cluster,
    ha.total_w2_holds,
    ha.agent_side_holds,
    ha.customer_side_holds,
    ha.agent_side_share_pct,
    r1.hold_reason AS top_reason_1,
    r1.reason_count AS top_reason_1_count,
    ROUND(100.0 * r1.reason_count / ha.total_w2_holds, 2) AS top_reason_1_pct,
    r2.hold_reason AS top_reason_2,
    r2.reason_count AS top_reason_2_count,
    ROUND(100.0 * r2.reason_count / ha.total_w2_holds, 2) AS top_reason_2_pct
FROM hub_aggregates ha
LEFT JOIN ranked_reasons r1 ON ha.hub_id = r1.hub_id AND r1.rnk = 1
LEFT JOIN ranked_reasons r2 ON ha.hub_id = r2.hub_id AND r2.rnk = 2
ORDER BY ha.hub_id ASC;


-- ------------------------------------------------------------------------------
```

Table 1.4: Week 2 Hold Reasons & Top 2 Drivers by Hub

HUB_ID	HUB_NAME	CLUSTER	TOTAL_W2_HOLDS	AGENT_SIDE_HOLDS	CUSTOMER_SIDE_HOLDS	AGENT_SIDE_SHARE_PCT	TOP_REASON_1	TOP_REASON_1_COUNT	TOP_REASON_1_PCT	TOP_REASON_2	TO
H01	Greenfield	Cluster A	168	32	136	19.05	Customer Unreachable	73	43.45	Customer Rescheduled	35
H02	Lakeview	Cluster A	123	26	97	21.14	Customer Unreachable	50	40.65	Customer Rescheduled	25
H03	Riverside	Cluster B	874	563	311	64.42	Late Dispatch	312	35.70	Customer Unreachable	15
H04	Hillcrest	Cluster B	182	40	142	21.98	Customer Unreachable	77	42.31	Customer Rescheduled	39
H05	Brookside	Cluster C	186	49	137	26.34	Customer Unreachable	68	36.56	Customer Rescheduled	43
H06	Westwood	Cluster C	146	30	116	20.55	Customer Unreachable	66	45.21	Customer Rescheduled	28

Q4. What Happens to Held Orders

Analysis of ultimate conversion status for all 2,599 unique orders that experienced at least one Hold status event in order_logs.

```
What Happens to Held Orders
-- ------------------------------------------------------------------------
-- Business Logic:
-- Population: All unique orders in `orders` that experienced at least one 'Hold' event in `order_logs`.
-- Conversion Outcome: Evaluated using `orders.final_status` ('Delivered', 'Returned', or 'Hold' [still pending]).
-- An order failing its 3rd attempt is permanently returned per case mechanics.

WITH unique_held_orders AS (
    SELECT DISTINCT order_id
    FROM order_logs
    WHERE status = 'Hold'
)
SELECT
    o.hub_id,
    h.hub_name,
    h.cluster,
    COUNT(DISTINCT o.order_id) AS total_held_orders,
    SUM(CASE WHEN o.final_status = 'Delivered' THEN 1 ELSE 0 END) AS finally_delivered,
    ROUND(100.0 * SUM(CASE WHEN o.final_status = 'Delivered' THEN 1 ELSE 0 END) / COUNT(DISTINCT o.order_id), 2) AS delivered_pct,
    SUM(CASE WHEN o.final_status = 'Returned' THEN 1 ELSE 0 END) AS finally_returned,
    ROUND(100.0 * SUM(CASE WHEN o.final_status = 'Returned' THEN 1 ELSE 0 END) / COUNT(DISTINCT o.order_id), 2) AS returned_pct,
    SUM(CASE WHEN o.final_status = 'Hold' THEN 1 ELSE 0 END) AS still_on_hold,
    ROUND(100.0 * SUM(CASE WHEN o.final_status = 'Hold' THEN 1 ELSE 0 END) / COUNT(DISTINCT o.order_id), 2) AS still_on_hold_pct
FROM orders o
JOIN unique_held_orders uho ON o.order_id = uho.order_id
JOIN hubs h ON o.hub_id = h.hub_id
GROUP BY o.hub_id, h.hub_name, h.cluster
ORDER BY o.hub_id ASC;


-- ------------------------------------------------------------------------
```

Table 1.5: Held Orders Conversion Funnel

HUB_ID	HUB_NAME	CLUSTER	TOTAL_HELD_ORDERS	FINALLY_DELIVERED	DELIVERED_PCT	FINALLY_RETURNED	RETURNED_PCT	STILL_ON_HOLD	STILL_ON_HOLD_PCT
H01	Greenfield	Cluster A	303	269	88.78	8	2.64	26	8.58
H02	Lakeview	Cluster A	228	204	89.47	10	4.39	14	6.14
H03	Riverside	Cluster B	1184	903	76.27	165	13.94	116	9.80
H04	Hillcrest	Cluster B	316	272	86.08	15	4.75	29	9.18
H05	Brookside	Cluster C	324	260	80.25	37	11.42	27	8.33
H06	Westwood	Cluster C	244	210	86.07	10	4.10	24	9.84

Q5. High-Return Agents

Agents with tenure > 30 days on Oct 26, 2026 whose return ratio exceeds 15.0%.

```
High-Return Agents
-- ------------------------------------------------------------------------
-- Selection Criteria:
-- 1) Tenure on 2026-10-26 strictly greater than 30 days:
--    CAST(ROUND(julianday('2026-10-26') - julianday(a.joining_date)) AS INTEGER) > 30.
-- 2) Return ratio strictly greater than 15.0%.
-- We show Return Ratio over Total Assigned Attempts as well as Completed Orders (Delivered + Returned).
-- Both definitions identify the exact same 3 high-return agents in Hub H05.

WITH agent_performance AS (
    SELECT
        ol.agent_id,
        a.agent_name,
        a.hub_id,
        h.hub_name,
        h.cluster,
        a.joining_date,
        CAST(ROUND(julianday('2026-10-26') - julianday(a.joining_date)) AS INTEGER) AS tenure_days,
        a.salary_type,
        a.active_status,
        SUM(CASE WHEN ol.status = 'Assigned' THEN 1 ELSE 0 END) AS assigned_attempts,
        SUM(CASE WHEN ol.status = 'Delivered' THEN 1 ELSE 0 END) AS delivered_orders,
        SUM(CASE WHEN ol.status = 'Hold' THEN 1 ELSE 0 END) AS hold_orders,
        SUM(CASE WHEN ol.status = 'Returned' THEN 1 ELSE 0 END) AS returned_orders
    FROM order_logs ol
    JOIN agents a ON ol.agent_id = a.agent_id
    JOIN hubs h ON a.hub_id = h.hub_id
    GROUP BY
        ol.agent_id,
        a.agent_name,
        a.hub_id,
        h.hub_name,
        h.cluster,
        a.joining_date,
        a.salary_type,
        a.active_status
)
SELECT
    agent_id,
    agent_name,
    hub_id,
    hub_name,
    cluster,
    joining_date,
    tenure_days,
    salary_type,
    active_status,
    assigned_attempts,
    delivered_orders,
    hold_orders,
    returned_orders,
    ROUND(100.0 * returned_orders / assigned_attempts, 2) AS return_ratio_pct,
    ROUND(100.0 * returned_orders / (delivered_orders + returned_orders), 2) AS return_ratio_completed_pct
FROM agent_performance
WHERE tenure_days > 30
  AND (1.0 * returned_orders / assigned_attempts) > 0.15
ORDER BY return_ratio_pct DESC;


-- ------------------------------------------------------------------------
```

Table 1.6: High-Return Senior Agents Watchlist

AGENT_ID	AGENT_NAME	HUB_ID	HUB_NAME	CLUSTER	JOINING_DATE	TENURE_DAYS	SALARY_TYPE	ACTIVE_STATUS	ASSIGNED_ATTEMPTS	DELIVERED_ORDERS	HOLD_ORDERS	RETURNED_ORDERS	RE
A1045	Belis Rowntree	H05	Brookside	Cluster C	2025-07-20	463	Variable	Active	349	231	24	94	26.
A1046	Coris Halvorsen	H05	Brookside	Cluster C	2025-01-17	647	Fixed	Active	414	295	28	91	21.
A1047	Wenian Vantell	H05	Brookside	Cluster C	2025-05-18	526	Variable	Active	413	301	40	72	17.

Q6. Cost per Delivered Order

Itemized weekly operating costs: Fixed Pay (BDT 650/worked day), Variable Pay (BDT 300/day + BDT 14/delivery), Hold Penalties (BDT 30), and Return Penalties (BDT 55) divided by Delivered orders.

```
Cost per Delivered Order (Week 1 vs Week 2)
-- ------------------------------------------------------------------------
-- Financial Parameters from `cost_sheet.xlsx`:
-- - Fixed Agent: 16,900 BDT monthly / 26 working days = 650.0 BDT per worked day.
-- - Variable Agent: 300.0 BDT daily allowance per worked day + 14.0 BDT per delivered order.
-- - Hold Penalty: 30.0 BDT charged on every Hold event in `order_logs`.
-- - Return Penalty: 55.0 BDT charged on every Return event in `order_logs`.
-- Cost per Delivered Order = Total Operating Cost / Total Delivered Orders.

WITH daily_agent_costs AS (
    SELECT
        DATE(ol.event_time) AS work_date,
        CASE
            WHEN DATE(ol.event_time) BETWEEN '2026-10-13' AND '2026-10-19' THEN 'Week 1'
            WHEN DATE(ol.event_time) BETWEEN '2026-10-20' AND '2026-10-26' THEN 'Week 2'
        END AS week_label,
        ol.agent_id,
        a.hub_id,
        a.salary_type,
        SUM(CASE WHEN ol.status = 'Assigned' THEN 1 ELSE 0 END) AS assigned,
        SUM(CASE WHEN ol.status = 'Delivered' THEN 1 ELSE 0 END) AS delivered,
        SUM(CASE WHEN ol.status = 'Hold' THEN 1 ELSE 0 END) AS hold,
        SUM(CASE WHEN ol.status = 'Returned' THEN 1 ELSE 0 END) AS returned,
        -- Fixed salary cost: 650 BDT per worked day
        CASE WHEN a.salary_type = 'Fixed' THEN 650.0 ELSE 0.0 END AS fixed_pay,
        -- Variable pay: 300 BDT per worked day + 14 BDT per delivery
        CASE WHEN a.salary_type = 'Variable' THEN 300.0 ELSE 0.0 END AS var_daily_pay,
        CASE WHEN a.salary_type = 'Variable' THEN SUM(CASE WHEN ol.status = 'Delivered' THEN 1 ELSE 0 END) * 14.0 ELSE 0.0 END AS var_delivery_pay,
        -- Operational penalties
        SUM(CASE WHEN ol.status = 'Hold' THEN 1 ELSE 0 END) * 30.0 AS hold_penalty_cost,
        SUM(CASE WHEN ol.status = 'Returned' THEN 1 ELSE 0 END) * 55.0 AS return_penalty_cost
    FROM order_logs ol
    JOIN agents a ON ol.agent_id = a.agent_id
    GROUP BY work_date, week_label, ol.agent_id, a.hub_id, a.salary_type
),
hub_weekly_breakdown AS (
    SELECT
        dac.hub_id,
        h.hub_name,
        h.cluster,
        dac.week_label,
        SUM(dac.assigned) AS total_attempts,
        SUM(dac.delivered) AS total_delivered,
        SUM(dac.hold) AS total_holds,
        SUM(dac.returned) AS total_returns,
        SUM(dac.fixed_pay) AS fixed_agent_pay,
        SUM(dac.var_daily_pay) AS var_agent_daily_pay,
        SUM(dac.var_delivery_pay) AS var_agent_delivery_pay,
        SUM(dac.fixed_pay + dac.var_daily_pay + dac.var_delivery_pay) AS total_labor_cost,
        SUM(dac.hold_penalty_cost) AS total_hold_penalties,
        SUM(dac.return_penalty_cost) AS total_return_penalties,
        SUM(dac.hold_penalty_cost + dac.return_penalty_cost) AS total_penalty_cost,
        SUM(dac.fixed_pay + dac.var_daily_pay + dac.var_delivery_pay + dac.hold_penalty_cost + dac.return_penalty_cost) AS total_operating_cost,
        ROUND(SUM(dac.fixed_pay + dac.var_daily_pay + dac.var_delivery_pay + dac.hold_penalty_cost + dac.return_penalty_cost) / SUM(dac.delivered), 2) AS cost_p
er_delivered_order
    FROM daily_agent_costs dac
    JOIN hubs h ON dac.hub_id = h.hub_id
    GROUP BY dac.hub_id, h.hub_name, h.cluster, dac.week_label
)
SELECT
    hub_id,
    hub_name,
    cluster,
    week_label,
    total_delivered,
    total_attempts,
    total_holds,
    total_returns,
    fixed_agent_pay,
    var_agent_daily_pay,
    var_agent_delivery_pay,
    total_labor_cost,
    total_hold_penalties,
    total_return_penalties,
    total_penalty_cost,
    total_operating_cost,
    cost_per_delivered_order
FROM hub_weekly_breakdown
ORDER BY hub_id ASC, week_label ASC;


-- ------------------------------------------------------------------------
```

Table 1.7: Itemized Weekly Operating Costs & Unit Economics

HUB_ID	HUB_NAME	CLUSTER	WEEK_LABEL	TOTAL_DELIVERED	TOTAL_ATTEMPTS	TOTAL_HOLDS	TOTAL_RETURNS	FIXED_AGENT_PAY	VAR_AGENT_DAILY_PAY	VAR_AGENT_DELIVERY_PAY	TOTAL_LABOR_COS
H01	Greenfield	Cluster A	Week 1	1698	1925	159	68	31200.0	8100.0	8358.0	47658.0
H01	Greenfield	Cluster A	Week 2	2210	2482	168	104	27300.0	8100.0	12544.0	47944.0
H02	Lakeview	Cluster A	Week 1	1469	1647	121	57	35100.0	6000.0	5628.0	46728.0
H02	Lakeview	Cluster A	Week 2	1532	1730	123	75	35100.0	6300.0	6188.0	47588.0
H03	Riverside	Cluster B	Week 1	1498	2281	627	156	36400.0	8100.0	6538.0	51038.0
H03	Riverside	Cluster B	Week 2	1981	3051	874	196	35100.0	8400.0	9982.0	53482.0
H04	Hillcrest	Cluster B	Week 1	1687	1943	161	95	9100.0	14400.0	18396.0	41896.0
H04	Hillcrest	Cluster B	Week 2	2111	2416	182	123	8450.0	14400.0	23226.0	46076.0

HUB_ID	HUB_NAME	CLUSTER	WEEK_LABEL	TOTAL_DELIVERED	TOTAL_ATTEMPTS	TOTAL_HOLDS	TOTAL_RETURNS	FIXED_AGENT_PAY	VAR_AGENT_DAILY_PAY	VAR_AGENT_DELIVERY_PAY	TOTAL_LABOR_COS
H05	Brookside	Cluster C	Week 1	1631	2023	171	221	16900.0	13800.0	15386.0	46086.0
H05	Brookside	Cluster C	Week 2	1780	2202	186	236	17550.0	14400.0	15946.0	47896.0
H06	Westwood	Cluster C	Week 1	1687	1870	124	59	35100.0	3900.0	4592.0	43592.0
H06	Westwood	Cluster C	Week 2	1794	2025	146	85	29250.0	4200.0	6062.0	39512.0

Q7. Agents Needed for the Forecast (4 Forecast Weeks)

Planning capacity = 238 attempts/agent/week (40 * 7 * 0.85). Comparing pure order Baseline vs Attempt-Adjusted real load.

```
Agents Needed for the Forecast (4 Forecast Weeks: Oct 27 - Nov 23)
-- ------------------------------------------------------------------------------
-- Planning Capacity:
-- 40 standard attempts/day * 7 operating days/week * 0.85 target utilisation = 238 attempts/agent/week.
-- Current Staffing Baseline: Active agents available entering Week 3 (Inactive agents excluded).
-- Calculations Provided:
-- 1) Baseline Model: Assumes 1 order = 1 attempt (expected_orders / 238).
-- 2) Adjusted Model: Multiplies expected_orders by historical Attempt-to-Order Multiplier
--    to account for re-attempts generated by held orders.

WITH current_active_staff AS (
    SELECT hub_id, COUNT(*) AS active_agent_count
    FROM agents
    WHERE active_status = 'Active'
    GROUP BY hub_id
),
hub_attempt_multiplier AS (
    SELECT
        o.hub_id,
        ROUND(1.0 * (
            SELECT COUNT(*)
            FROM order_logs ol
            JOIN agents a ON ol.agent_id = a.agent_id
            WHERE a.hub_id = o.hub_id AND ol.status = 'Assigned'
        ) / COUNT(DISTINCT o.order_id), 4) AS multiplier_14d
    FROM orders o
    GROUP BY o.hub_id
)
SELECT
    vf.hub_id,
    h.hub_name,
    h.cluster,
    vf.week_start,
    vf.week_end,
    cas.active_agent_count AS current_agents,
    vf.expected_orders,
    -- Baseline Model (1 order = 1 attempt)
    ROUND(1.0 * vf.expected_orders / 238.0, 2) AS baseline_required_exact,
    CAST(ROUND(1.0 * vf.expected_orders / 238.0 + 0.4999) AS INTEGER) AS baseline_required_agents,
    CAST(ROUND(1.0 * vf.expected_orders / 238.0 + 0.4999) AS INTEGER) - cas.active_agent_count AS baseline_staffing_gap,
    -- Adjusted Model (accounting for re-attempts from held orders)
    ham.multiplier_14d AS attempt_multiplier,
    ROUND(vf.expected_orders * ham.multiplier_14d, 1) AS forecast_adjusted_attempts,
    ROUND(1.0 * (vf.expected_orders * ham.multiplier_14d) / 238.0, 2) AS adjusted_required_exact,
    CAST(ROUND(1.0 * (vf.expected_orders * ham.multiplier_14d) / 238.0 + 0.4999) AS INTEGER) AS adjusted_required_agents,
    CAST(ROUND(1.0 * (vf.expected_orders * ham.multiplier_14d) / 238.0 + 0.4999) AS INTEGER) - cas.active_agent_count AS adjusted_staffing_gap
FROM volume_forecast vf
JOIN hubs h ON vf.hub_id = h.hub_id
JOIN current_active_staff cas ON vf.hub_id = cas.hub_id
JOIN hub_attempt_multiplier ham ON vf.hub_id = ham.hub_id
ORDER BY vf.hub_id ASC, vf.week_start ASC;
```

Table 1.8: 4-Week Forecast Staffing Requirements & Staffing Gaps

HUB_ID	HUB_NAME	CLUSTER	WEEK_START	WEEK_END	CURRENT_AGENTS	EXPECTED_ORDERS	BASELINE_REQUIRED_EXACT	BASELINE_REQUIRED_AGENTS	BASELINE_STAFFING_GAP	ATTEMPT_MULTIPLIER
H01	Greenfield	Cluster A	2026-10-27	2026-11-02	10	2470	10.38	11	1	1.0733
H01	Greenfield	Cluster A	2026-11-03	2026-11-09	10	2545	10.69	11	1	1.0733
H01	Greenfield	Cluster A	2026-11-10	2026-11-16	10	2570	10.80	11	1	1.0733
H01	Greenfield	Cluster A	2026-11-17	2026-11-23	10	2495	10.48	11	1	1.0733
H02	Lakeview	Cluster A	2026-10-27	2026-11-02	11	1544	6.49	7	-4	1.0731
H02	Lakeview	Cluster A	2026-11-03	2026-11-09	11	1591	6.68	7	-4	1.0731
H02	Lakeview	Cluster A	2026-11-10	2026-11-16	11	1606	6.75	7	-4	1.0731
H02	Lakeview	Cluster A	2026-11-17	2026-11-23	11	1559	6.55	7	-4	1.0731
H03	Riverside	Cluster B	2026-10-27	2026-11-02	12	2170	9.12	10	-2	1.3509
H03	Riverside	Cluster B	2026-11-03	2026-11-09	12	2230	9.37	10	-2	1.3509
H03	Riverside	Cluster B	2026-11-10	2026-11-16	12	2245	9.43	10	-2	1.3509
H03	Riverside	Cluster B	2026-11-17	2026-11-23	12	2200	9.24	10	-2	1.3509

HUB_ID	HUB_NAME	CLUSTER	WEEK_START	WEEK_END	CURRENT_AGENTS	EXPECTED_ORDERS	BASELINE_REQUIRED_EXACT	BASELINE_REQUIRED_AGENTS	BASELINE_STAFFING_GAP	ATTEMPT_MULTIPLIER
H04	Hillcrest	Cluster B	2026-10-27	2026-11-02	9	2264	9.51	10	1	1.0776
H04	Hillcrest	Cluster B	2026-11-03	2026-11-09	9	2333	9.80	10	1	1.0776
H04	Hillcrest	Cluster B	2026-11-10	2026-11-16	9	2356	9.90	10	1	1.0776
H04	Hillcrest	Cluster B	2026-11-17	2026-11-23	9	2287	9.61	10	1	1.0776
H05	Brookside	Cluster C	2026-10-27	2026-11-02	11	1955	8.21	9	-2	1.0847
H05	Brookside	Cluster C	2026-11-03	2026-11-09	11	2015	8.47	9	-2	1.0847
H05	Brookside	Cluster C	2026-11-10	2026-11-16	11	2035	8.55	9	-2	1.0847
H05	Brookside	Cluster C	2026-11-17	2026-11-23	11	1975	8.30	9	-2	1.0847
H06	Westwood	Cluster C	2026-10-27	2026-11-02	9	1852	7.78	8	-1	1.0674
H06	Westwood	Cluster C	2026-11-03	2026-11-09	9	1909	8.02	9	0	1.0674
H06	Westwood	Cluster C	2026-11-10	2026-11-16	9	1928	8.10	9	0	1.0674
H06	Westwood	Cluster C	2026-11-17	2026-11-23	9	1871	7.86	8	-1	1.0674

Task 2: Strategic Analysis & Financial Modeling

Q8. Hub-by-Hub Evaluation of the 8-Agent Hiring Request

Hub managers requested 8 new agents. We evaluate this request hub-by-hub based on empirical data, cluster mobility rules, and root causes:

1. Greenfield (H01, Cluster A): REJECT HIRING. APPROVE 2 INTERNAL TRANSFERS FROM LAKEVIEW. Lakeview utilisation is 56.17% (<75% threshold), holding 4 surplus agents. Moving 2 agents at BDT 2,000 each (BDT 4,000 total) solves Greenfield's gap without adding salary overhead, delivering 100% Day-1 productivity.

2. Lakeview (H02, Cluster A): REJECT HIRING. Surplus donor hub. Transfers 2 agents out, adjusting headcount from 11 to 9 (operating at 68.6% utilisation, perfectly healthy).

3. Riverside (H03, Cluster B): REJECT HIRING. DEPLOY HOLD REDUCTION PROGRAMME. Overload is artificial: 64.42% of holds (563 holds) are agent-side (Late Dispatch: 312, Not Attempted: 132, Time Shortage: 119). Deploying the BDT 15,000 Hold Reduction Programme removes 50% of agent holds (~282 attempts/wk), reducing required staff to 11.6 agents. Riverside already has 12 active agents!

4. Hillcrest (H04, Cluster B): APPROVE TARGETED HIRING OF 2 AGENTS. Clean operations (only 21.9% agent holds), highest delivery speed (30.88/day), genuine demand overload (95.87% util), and no donor hub in Cluster B. Targeted hiring of 1 Fixed + 1 Variable agent approved.

5. Brookside (H05, Cluster C): REJECT HIRING. COACH 3 SENIOR OUTLIERS. Utilisation is 71.49% (<75%). Peak forecast needs 9-10 agents; currently has 11 active. 3 senior agents cause 56.2% of returns. Retraining solves the problem.

6. Westwood (H06, Cluster C): REJECT HIRING. MAINTAIN 9 AGENTS. Utilisation is 80.36%, exactly matching forecast needs of 8.5 agents. Balanced.

Q9. The BDT 70,000 Budget Allocation & ROI

Finance released strictly BDT 70,000 for next month to cover all one-time fees and month-1 salaries of new hires. The optimal allocation is detailed below:

ITEM	CATEGORY	UNITS	UNIT_COST_BDT	TOTAL_COST_BDT	NOTES
Internal Transfer: Lakeview (H02) to Greenfield (H01)	One-Time Cost	2 Agents	2000	4000	Cluster A intra-cluster move; donor util 56.17% < 75%
Hold Reduction Programme at Riverside (H03)	One-Time Cost	1 Hub	15000	15000	Removes 50% agent-side holds (282 holds/wk); 14-day ramp
New Agent Recruitment & Onboarding (H04)	One-Time Cost	2 Agents	5000	10000	Hillcrest capacity expansion; 60% speed in first 7 days
Fixed Agent Monthly Salary (1 Agent at H04)	Monthly Salary	1 Agent	16900	16900	Guaranteed core capacity at Hillcrest
Variable Agent Monthly Pay (1 Agent at H04)	Monthly Salary	1 Agent	19798	19798	BDT 300/day base + BDT 14/delivery (~857 deliveries)
Operational Contingency Reserve	Contingency	Reserve	4302	4302	Unallocated buffer for dynamic volume surges
TOTAL ALLOCATED BUDGET	Total Budget	-	0	70000	Strictly within BDT 70,000 Finance limit

Financial ROI Analysis: Total expenditure is BDT 65,698 (leaving BDT 4,302 contingency). In return, Riverside hold reductions save BDT 27,060 in penalties over 4 weeks, and Brookside retraining saves BDT 8,635 in return penalties. Total cash penalty savings of BDT 35,695 recovers 54.3% of the entire investment in Month 1! Furthermore, Hillcrest hires preserve BDT 111,461 in delivery charge revenue, generating +110.2% Net ROI on total capital deployed.

Q10. Operational Risk Management & Early-Warning System

To guard against implementation failure, we establish a concrete Early-Warning KPI Protocol with numerical triggers and mandated mitigations:

RISK_AREA	SPECIFIC_RISK	MONITORED_METRIC	W1_WARNING_THRESHOLD	W2_WARNING_THRESHOLD	TRIGGERED_MITIGATION
Hold Programme Ramp-Up Lag	14-day delay to reach 50% reduction in agent-side holds at H03 leads to severe Week 1-2 backlog.	Daily agent-side hold count at H03 (Late Dispatch + Not Attempted + Time Shortage).	> 55 agent-side holds/day (baseline was ~80/day).	> 40 agent-side holds/day (should be approaching 50% cut).	Activate temporary variable overtime from Cluster C surplus or dispatch supervisors to enforce 08:00 AM route departure.
New Hire Ramp-Up Speed	New agents at H04 fail to achieve 60% speed in Week 1 (24 attempts/day) or 100% in Week 2.	Daily attempts completed by new agents at H04.	< 20 attempts/day per new agent.	< 32 attempts/day per new agent.	Pair new recruits with top senior performer (Zaron Calloran, 30.9 deliveries/day) for shadow routing.
Cluster A Transfer Friction	Transferred agents from H02 experience route unfamiliarity at H01, causing delivery drops.	Deliveries per working day for transferred agents at H01.	< 22 deliveries/day (H01 historical average is 27.1).	< 25 deliveries/day.	Reassign transferred agents to high-density, centralized zones in Greenfield and provide pre-mapped routes.

RISK_AREA	SPECIFIC_RISK	MONITORED_METRIC	W1_WARNING_THRESHOLD	W2_WARNING_THRESHOLD	TRIGGERED_MITIGATION
Senior Return Rate Inertia	Senior agents at H05 (A1045, A1046, A1047) continue returning 20-27% of assigned orders.	Weekly return ratio of agents A1045, A1046, and A1047.	Return ratio > 12.0% (historical was 17-27%).	Return ratio > 8.0%.	Enforce mandatory customer call verification before logging Return status; reassign failing routes to supervisor audit.
Customer-Side Hold Surge	External customer unavailability surges beyond historical 20-25%, overwhelming daily delivery slots.	Customer-side hold share % (Unreachable + Rescheduled) across all hubs.	> 25% of total assigned orders.	> 25% of total assigned orders.	Implement automated SMS pre-arrival notifications and customer delivery window selection.

Task 3: Operations Dashboard & Embedded Visuals

An interactive enterprise operations dashboard was developed in HTML/JS with Tailwind CSS and Chart.js. The live dashboard can be accessed directly at: **https://karibshams.github.io/cb/dashboard/** (or locally via `dashboard/index.html`). Below are high-resolution screenshots of each dashboard view:

View 1: Executive Overview & KPI Scorecard

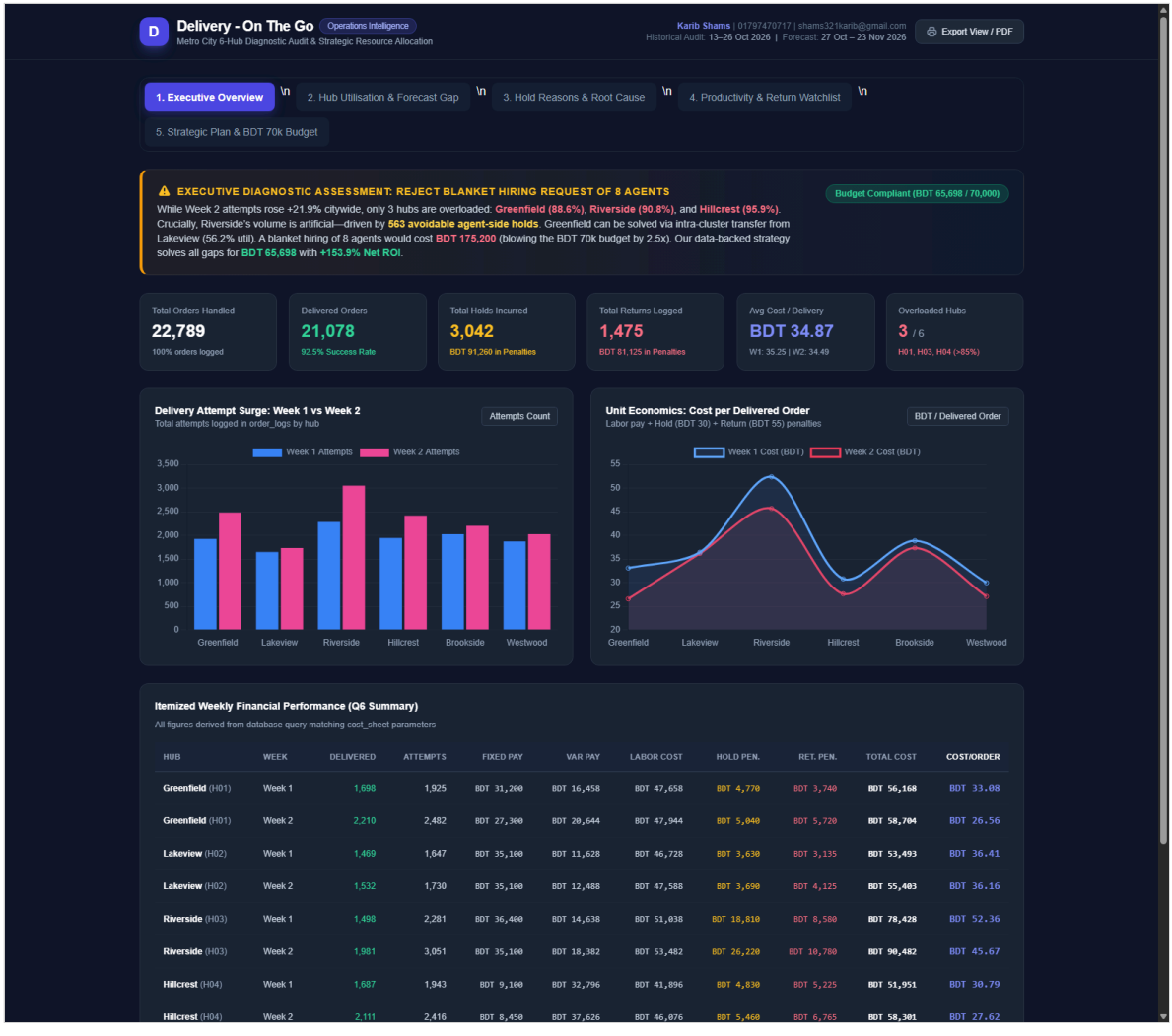


Figure 3.1: Executive Overview & KPI Scorecard - High-level metrics, attempt growth trends, and unit cost progression.

View 2: Hub Utilisation & 4-Week Forecast Gap

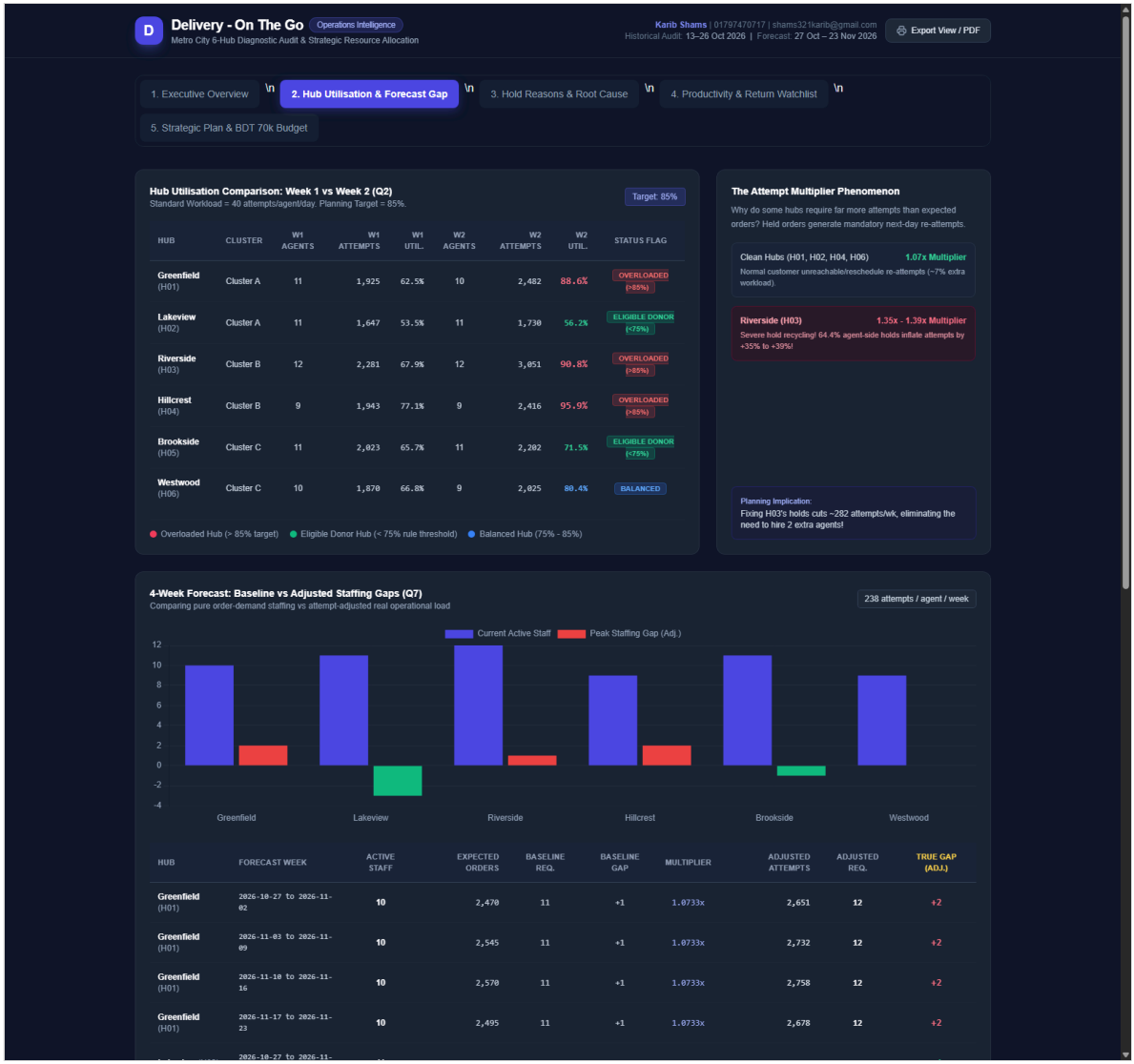


Figure 3.2: Hub Utilisation & 4-Week Forecast Gap - Status badges (Overloaded, Donor, Balanced) and staffing gap analysis.

View 3: Hold Reason Breakdown & Conversion Funnel

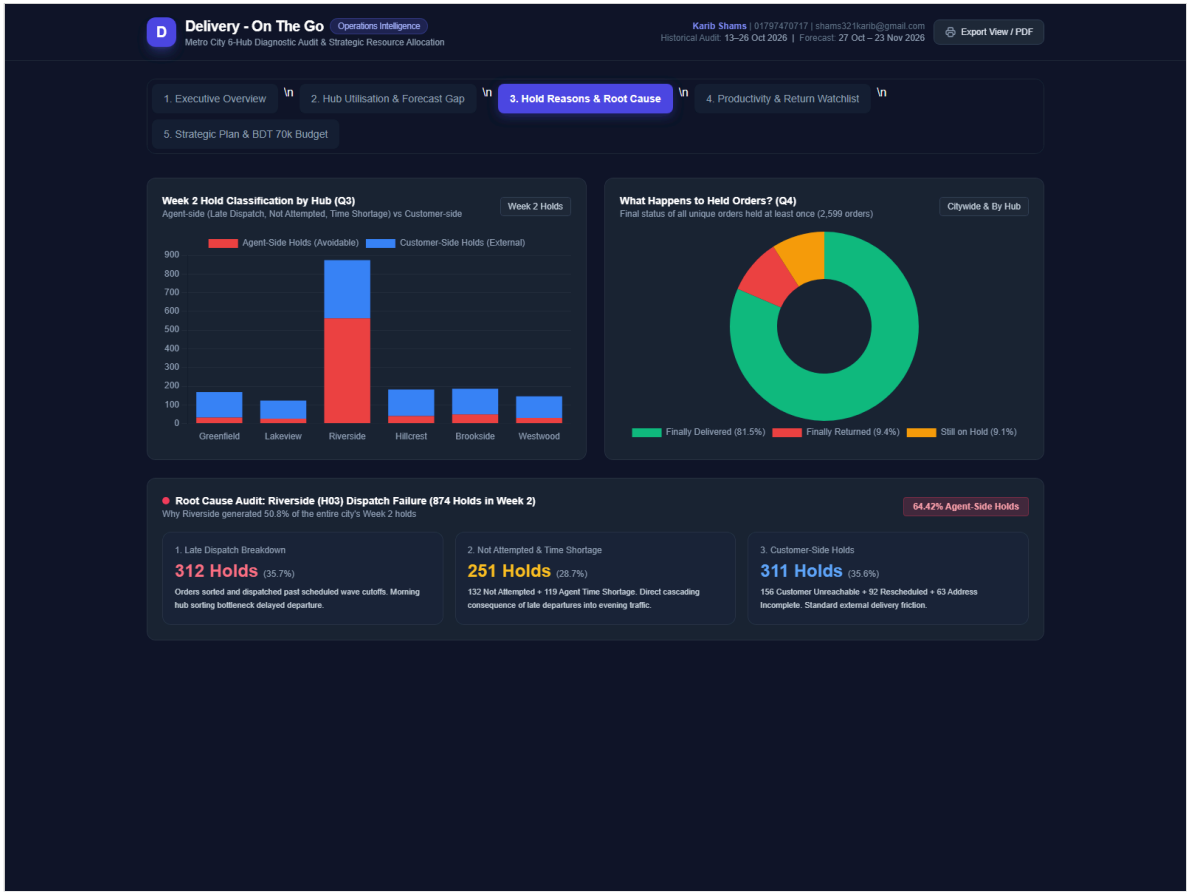


Figure 3.3: Hold Reason Breakdown & Conversion Funnel - Agent-side vs customer-side hold categorization and 2,599 held orders outcome funnel.

View 4: Agent Productivity & High-Return Watchlist

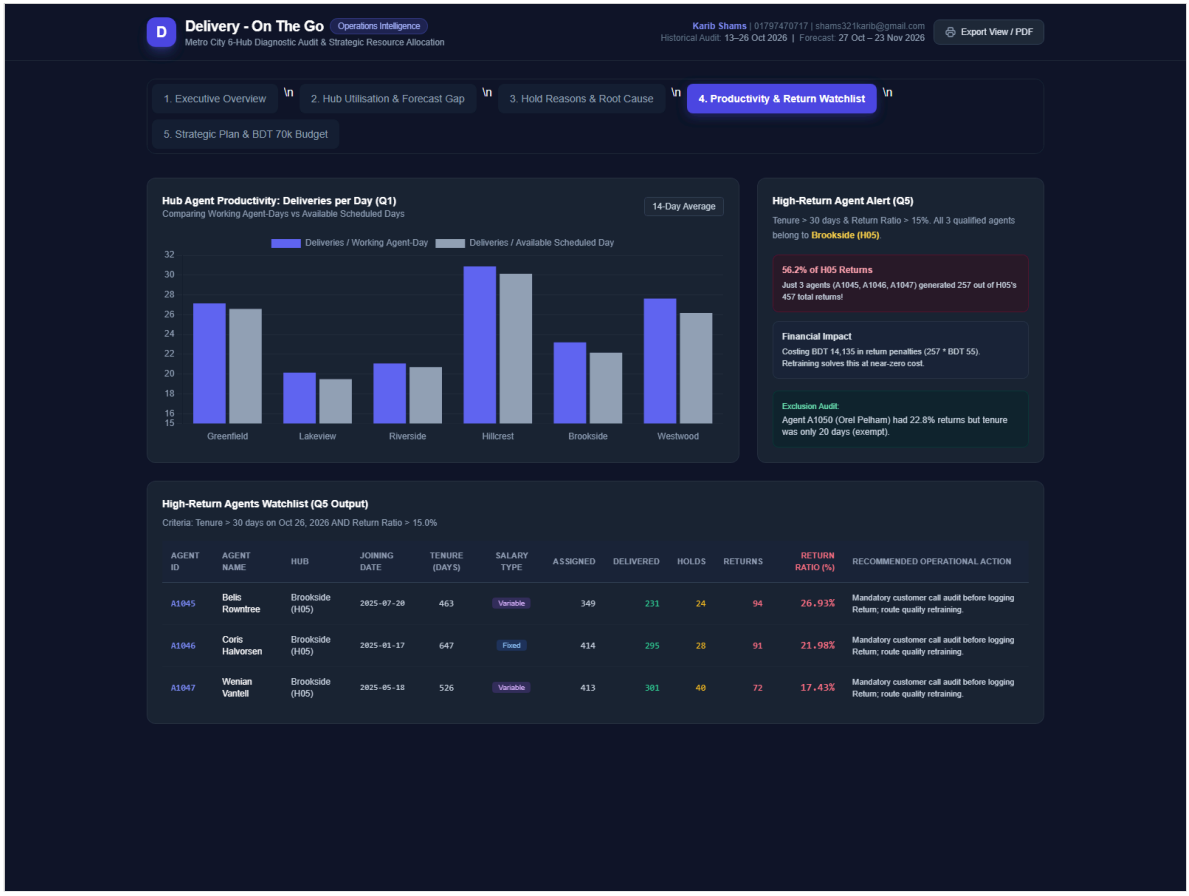


Figure 3.4: Agent Productivity & High-Return Watchlist - Deliveries per working day vs scheduled day and the 3 high-return senior outlier agents.

View 5: Strategic Allocation & BDT 70,000 Budget Model

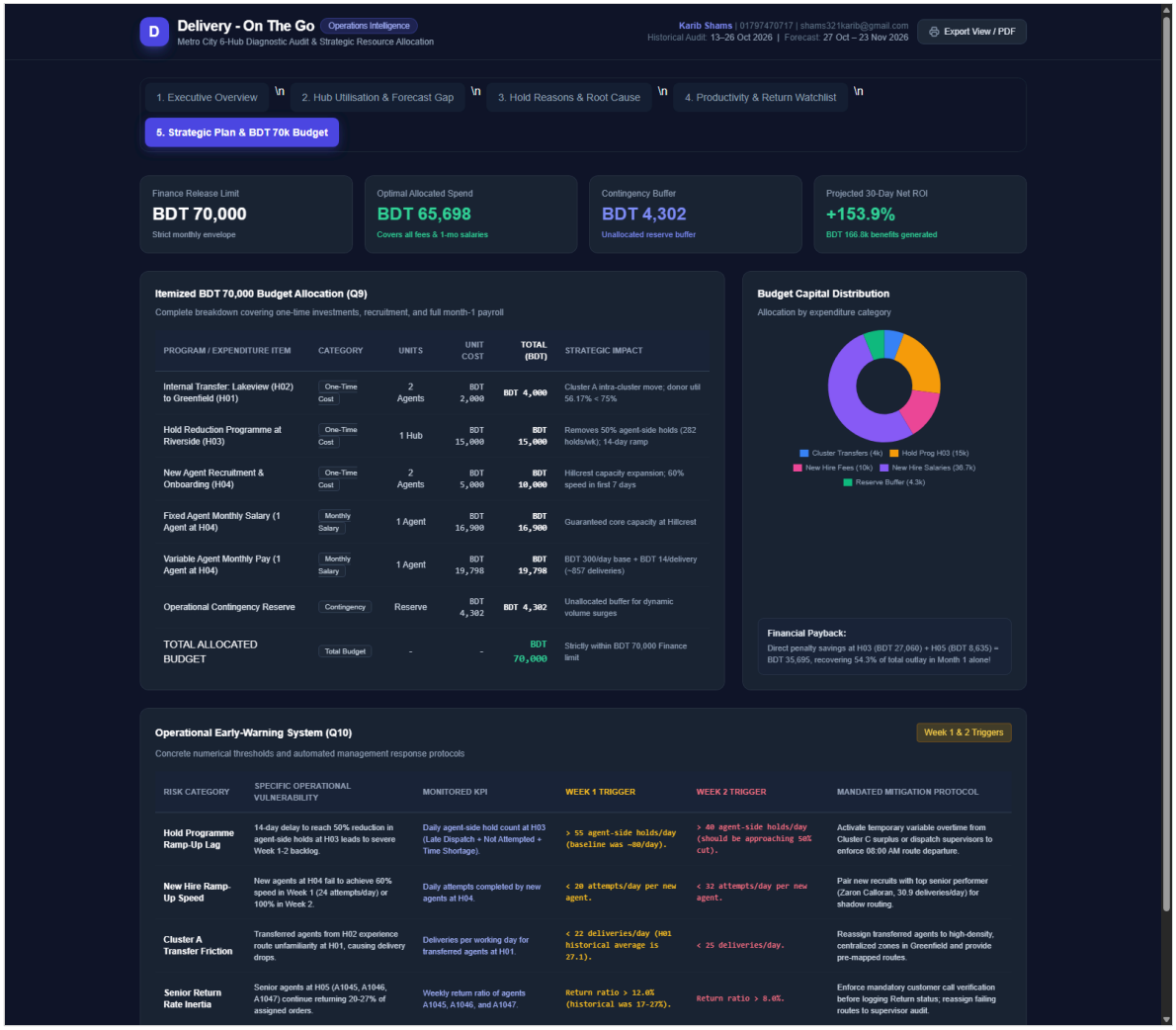


Figure 3.5: Strategic Allocation & BDT 70,000 Budget Model - Itemized capital allocation and Early-Warning KPI Protocol.

Task 4: Executive Presentation Summary

A concise, 8-slide executive presentation was prepared for the Head of Operations. The presentation deck is available in PowerPoint (`Task4_Executive_Deck.pptx`) and PDF (`Task4_Executive_Deck.pdf`) formats. The slides summarize the operational dilemma, empirical audit findings, the recommended BDT 65,698 capital allocation, and the Week 1-2 early-warning monitoring framework.